

سؤال

$$\frac{dX_t}{dt} = u_t(X_t) \quad \underbrace{X_0 \sim P_{init}} \quad \underbrace{X_1 \sim P_{data}} \\ 0 \leq t \leq 1$$

How to choose $u_t(\cdot)$ s.t. $X_1 \sim P_{data}$.

train $u_t^\theta(x)$ based on $Z^{(1)} \dots Z^{(n)} \sim P_{data}$

$u_t^{target}(x|z) \rightsquigarrow$ Def. a $u(\cdot)$ that when the above is run based off, results in $P_{t=1} = \delta(\cdot - z)$

Assumption

$$\rightarrow P_t(x|z) = \underline{\underline{N}}(\alpha_t z, \beta_t^2 I) \quad \begin{cases} \alpha_0 = 0, \beta_0 = 1 \\ \alpha_1 = 1, \beta_1 = 0 \end{cases}$$

$$u_t^{target}(x) = \int \underbrace{u_t^{target}(x|z)} \cdot \underbrace{P_t(x|z) \cdot P_{data}(z)}_{P_t(x)} dz.$$

requires

$$E \left\{ \left\| u_t^\theta(x) - \underbrace{u_t^{target}(x|z)} \right\|_2^2 \right\} = E \left\{ \left\| u_t^\theta(x) - \underbrace{u_t^{target}(x)}_{+C} \right\|_2^2 \right\}$$

Stochastic Diff. Eq.

$$\underline{\underline{dX_t}} = \underbrace{v_t(X_t) dt}_{\text{Deterministic Update}} + \underbrace{\sigma_t dW_t}_{\text{Diff. Wiener Process}} \quad \leftarrow \text{Stochastic Update}$$

$$W(t_i) - W(t_0) \sim \mathcal{N}(0, (t_i - t_0)^2) \\ W(t_i) - W(t_{i-1})$$

indep

$$W(t_{i+1}) - W(t_i)$$

Theorem: You want to generate X_t using an SDE

s.t. $P_{SDE}^t = P_{ODE}^t$. Now, what $V_t(\cdot)$ & σ_t
 $\forall t$

should be in order for this to happen.

$$dX_t = \underbrace{\left[u_t^{\text{target}}(X_t) + \frac{\sigma_t^2}{2} \nabla \log P_t(X_t) \right]}_{V_t(X_t)} dt + \sigma_t dW_t$$

Proof: P_t^{ODE} satisfies the continuity Eq:

$$\frac{\partial}{\partial t} P_t^{\text{ODE}} = -\text{div}(P_t u_t^{\text{target}}) - \underbrace{\frac{\sigma_t^2}{2} \Delta P_t(X_t)}_{\text{div}(\nabla P_t(X_t))} + \frac{\sigma_t^2}{2} \Delta P_t(X_t)$$

$$\text{div}: \Delta f(x_1, \dots, x_n) = \sum_{i=1}^n \frac{\partial^2 f}{\partial x_i^2} = \text{div}(\nabla f)$$

$$= -\text{div}\left(P_t u_t^{\text{target}} + \frac{\sigma_t^2}{2} \nabla P_t\right) + \frac{\sigma_t^2}{2} \Delta P_t(X_t)$$

$$P_t \left[u_t^{\text{target}} + \frac{\sigma_t^2}{2} \underbrace{\frac{\nabla P_t}{P_t}}_{\nabla \log P_t} \right]$$

$$\underbrace{\hspace{10em}}_{V_t}$$

$\Rightarrow$ if the SDE is of the form

$$dX_t = V_t(X_t) dt + \sigma_t dW_t \quad \text{then}$$

$$X_t \sim P_t^{\text{ODE}}$$

$$V_t(X_t) = u_t^{\text{target}}(X_t) + \frac{\sigma_t^2}{2} \nabla \log P_t \quad \propto$$

□

Fokker-Planck Lemma:

$$\varphi(\cdot) \in \mathcal{C}_c^\infty$$

Itô's lemma:

$$\mathbb{E} \left\{ \frac{d}{dt} \varphi(X_t) \right\} = \mathbb{E} \left\{ \langle \nabla \varphi(X_t), dX_t \rangle \right\} + \frac{1}{2} \mathbb{E} \left\{ \sum \frac{\partial^2 \varphi}{\partial x_i \partial x_j} d[X_t^i X_t^j] \right\}$$

$$d[X_t^i X_t^j] := \underbrace{dX_t^i dX_t^j}_{(dt)^2 = 0}$$

$$[u_t^i(X_t) dt + \sigma_t dW_t^i][u_t^j(X_t) dt + \sigma_t dW_t^j]$$

$$= u_t^i(X_t) u_t^j(X_t) (dt)^2 + u_t^i(X_t) \sigma_t dW_t^j dt + u_t^j(X_t) \sigma_t dW_t^i dt + \sigma_t^2 dW_t^i dW_t^j$$

$$dX_t = u_t dt + \sigma_t dW_t$$

$$+ u_t^j(X_t) \sigma_t dW_t^j dt + \sigma_t^2 dW_t^i dW_t^j$$

$$+ \sigma_t^2 dW_t^i dW_t^j$$

$$\mathbb{E} \left\{ \frac{d}{dt} \varphi(X_t) \right\} = \mathbb{E} \left\{ \langle \nabla \varphi(X_t), \frac{dX_t}{dt} \rangle \right\} + \mathbb{E} \left\{ \frac{\sigma_t^2}{2} \Delta \varphi(X_t) \right\}$$

$$\int \varphi(x) \frac{d}{dt} p_t(x) dx = \int \underbrace{\langle \nabla \varphi(x), u_t(x) \rangle}_{f \cdot F} \underbrace{p_t(x)}_{P_t(x)} dx + \int \frac{\sigma_t^2}{2} \Delta \varphi(x) \cdot p_t(x) dx$$

$$= \int \varphi(x) [g(x)] dx$$

scalar vector field

$$\text{div}(f \cdot F) = \langle \nabla f, F \rangle + f \cdot \text{div}(F)$$

$$A = \int \underbrace{\text{div}(\underbrace{f}_{\varphi} \cdot F)}_{\phi} - \int \underbrace{f}_{\phi} \cdot \text{div}(\underbrace{F}_{u_t}) = - \int \varphi(x) \cdot \text{div}(u_t p_t) dx$$

$$\operatorname{div}(f \cdot F) = \langle \nabla f, F \rangle + \underbrace{f \cdot \operatorname{div}(F)}_{\substack{\underbrace{\left(\frac{\sigma_t^2}{2} p_t\right)}_{\Delta \varphi} \quad \underbrace{\nabla \varphi}_{\text{B}}}}$$

$$\int_B dx = \underbrace{\int \operatorname{div}(f \cdot \widehat{F})}_{\circ} - \underbrace{\int \langle \nabla f, \widehat{F} \rangle}_{\substack{\text{C} \\ \nabla \varphi}} dx$$

$$\operatorname{div}(f \cdot F) = \langle \nabla f, \underbrace{\widehat{F}}_{\nabla \varphi} \rangle + f \cdot \operatorname{div}(F)$$

$$-\int_C dx = -\int \operatorname{div}(\underbrace{\nabla f}_{\circ} \cdot \varphi) + \int \varphi(x) \cdot \underbrace{\operatorname{div}(\nabla f^{\text{old}})}_{\frac{\sigma_t^2}{2} \Delta p_t} dx$$

$$\int \varphi(x) \frac{\partial p_t(x)}{\partial t} dx = \int \varphi(x) \operatorname{div}(p_t u_t)(x) dx + \int \varphi(x) \cdot \frac{\sigma_t^2}{2} \Delta p_t(x) dx$$

$$\Rightarrow \frac{\partial p}{\partial t}(x) = -\operatorname{div}(p_t u_t)(x) + \frac{\sigma_t^2}{2} \Delta p_t(x) \quad \square$$